portable low voltage UVC sterilising appliances. These are widely used in germicidal LEDs present in the Indian market for disinfection systems used for sterilisation of cabinets, luggage conveyor tunnels, robots, etc.

These 275nm UV LEDs emit light in 265nm to 275nm band at 10-17mJ/sec in 100mA drive. A lethal dosage of $0.6mJ/cm^2$ to $10mJ/cm^2$ is required to eliminate most pathogens. Besides being highly effective (thirty per cent more) than 254nm tubes that are used for sterilisation and disinfection, the UV LEDs are also helpful in the fight against Covid-19.

LEDchip Indus produces a full range of UVA, UVB, and UVC radiation as well as full-spectrum LEDs for specialty lighting applications.

LEDchip Indus
www.ledchipindus.com

Touch controllers

Microchip Technology has now extended its maXTouch portfolio with the inclusion of the new MXT288UD touch controller family.

MXT288UD-AM and MXT144UD-AM automotive-grade packaged touch screen controllers offer low-power mode, weatherproof operation, and glove touch detection in multi-function displays, touchpad and smart surfaces for vehicles, motorcycles, e-bikes, and car-sharing services. The devices enable secondary touch surfaces to be placed in both the interior of cars and the exterior of a motor vehicle such as handlebars, doors, electronic mirrors, control knobs, steering wheel, between seats, etc.

With the MXT288UD family's small 7mm × 7mm automotive-grade VQFN56 package, tier-one suppliers can now reduce board space by 75 per cent and greatly minimise the overall Bill of Materials (BoM) for these compact applications—all while exceeding the requirements for excellent and reliable touch performance.

Additionally, MXT288UD-AM and MXT144UD-AM devices enable detection and tracking of multi-finger thick gloves through a wide variety of overlay materials like leather, wood, or across uneven surfaces even in the presence of moisture.

Microchip
www.microchip.com

Power management IC

Empower Semiconductor, a provider of Integrated Voltage Regulators (IVR), has launched EP70xx, a cutting-edge series of power management ICs for significant energy savings in data centres. The device allows total integration of triple output DC-DC power supply into a single tiny 5mm × 5mm package with no external components.

This allows attaining up to 10x higher current density, 3x tighter accuracy during transients, and 1000x faster dynamic voltage. Having multiple power supplies in a single IC package reduces the concerns of component variation, sourcing, synchronisation, and stability. This simplifies the adoption of DC-DC converters by power designers who need multilevel voltage and current regulation.

The EP70xx family displays peak efficiencies of up to 91 per cent with a nearly flat efficiency curve up through 10A of output current. It showcases 1,000x faster dynamic voltage scaling, enabling fast and lossless processor state changes that can save 30 per cent or more of processor power.

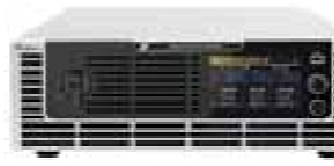
Empower Semiconductor
www.empowersemi.com

TEST & MEASUREMENT

High-power grid simulator

Chroma has added three new models: 61809, 61812, and 61815, to its 61800 series of regenerative grid simulators. These new products are four-quadrant, fully regenerative AC power sources having high power density and provide power ratings of 9kVA/12kVA/15kVA, respectively. The 3U height form factor allows the device to occupy less space and provide flexibility for users when configuring their system cabinet or test bench.

These devices can save operational costs. Additionally, users can extend the output power by connecting up to three units in parallel, thus achieving a high power density configuration of 45kVA output power at only 9U height.



Besides grid simulation for server power or electrical-commercial products, the new 61800 models can be applied to applications in green energy products, including PV inverters, energy storage systems (ESS), power conditioning systems (PCS), etc.

Chroma
www.croma.com

Oscilloscope

element14 and Avnet have added the new PicoScope 6000E series of USB oscilloscopes from Pico Technology to their portfolio. The new generation PicoScope 6000E series is an ideal measurement and analysis solution for the design and test engineers, OEMs working with high-performance embedded systems, and researchers and scientists working on multi-channel high-performance experiments in physics labs, particle accelerators, and similar facilities.

The PicoScope 6000E series consists of two models: PicoScope 6804E and PicoScope 6824E that also offer FlexRes. This specification reduces the sampling rate to enable high-resolution 10-bit or 12-bit resolution for audio and other analogue applications.

The PicoScope series also offers informative display and advanced waveform analysis, thereby increasing user productivity when conducting signal processing, high-speed communications, process control, avionics, power electronics, and automotive design.

Pico Technology
www.picotech.com